Princess Sumaya University for Technology
جامعة الأميرة سميّة للتكنولوجيا

PlantLine: A Plant disease image classifier

Project done in partial fulfillment of the Special topics in Data science and Artificial Intelligence course

- *Saif-aldein Ghassan Shoujen ---- 20240146*
- *Amer Abd-almoamen Mansour ---- 20240258*
- *Instructor: Dr.Tariq Bdair*

Problem Statement

Agriculture is one of the most important industries, feeding billions of people for thousands of years. One of the biggest threats facing this industry is plant disease, as it destroys crop yields yearly. This happens because most farmers aren't properly equipped and don't have access to the resources to combat these diseases.

This project aims to provide a fast, accessible, and easy to use tool to diagnose plants using images of their leaves. The crops targeted here are tomatoes, potatoes, and bell peppers as they are the most important and are very susceptible to fungal, bacterial ,and viral diseases.

Motivation

The motivation of this project is the lack of widespread implementation of computer vision in agriculture, especially in Jordan. Recent advancements enable the use of lightweight convolutional neural networks to accurately classify crop images.

Some motivation factors

- Food shortages and instability due to the spread of diseases
- The lack of expert pathologists in rural areas with limited resources
- The spread of smartphones and small devices among farmers, opening the way to image-based diagnosis
- The existence of labeled datasets with high data veracity
- Lightweight architecture

This project will help shine a light on the ways computer vision can be applied to agriculture to raise farm yields and save expenses and effort for farmers.

Dataset choice

For this task, this project uses the PlantVillage dataset[1], which is widely adopted In agricultural computer vision research. This dataset consists of roughly 20 thousand images spanning 15 classes and 3 crop species, captured under controlled conditions with uniform backgrounds.

[1] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in Plant Science, 7. https://doi.org/10.3389/fpls.2016.01419

Class structure

The class structure here is hierarchical in nature. Each crop species has multiple disease classes along with healthy ones as a control. For example: the tomato subset has early blight, late blight, leaf mold, septoria leaf spot, spider mites, target spot, mosaic virus, yellow leaf curl virus, and bacterial spot – along with healthy specimen.

Reason for Dataset selection

- Dataset is a publicly licensed repository for academic and research use.
- 20 thousand images provide enough data to prevent overfitting
- Prelabeled by pathology experts.

Data Preprocessing

- Resizing and rescaling: ensuring stable tensor operations and reduce memory overhead
- Normalization: will be normalized with mean and std of ImageNet as transfer learning will be used
- Train/Validation/Test split: 75% training, 25% testing, and 10% of the training used for validation

Data Augmentation

- Flipping and rotation: random horizontal/vertical flips and random rotation (30 degrees)
- Color Jitter and brightness adjustment: random changes to brightness, contrast, saturation, and hue.
- Cutout / Random erasing: to reduce reliance on localized texture patterns

Data ethics

- Licensing: Dataset is licensed for research and academic use, as well as the article being cited.
- Privacy: No personal information is included in the data
- Bias and fairness: It should be considered that the data is collected under a controlled environment not representative of real-world conditions. Augmentation is implemented to combat this
- Responsible deployment: this model must be framed as a decision support tool rather than a final diagnostic system. Farmers are encouraged to double check predictions , especially in low-confidence predictions.